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1-METHYL-4-PHENYLPYRIDINE (MPP^+) IS TOXIC TO MESENCEPHALIC DOPAMINE NEURONS IN CULTURE

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1-Methyl-4-phenylpyridine (MPP^+) is the product of oxidative metabolism by monoamine oxidase of the Parkinson-inducing agent 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP). MPP^+ was tested for neurotoxicity on the dopaminergic neurons of embryonic rat midbrain in culture. Compared with MPTP, MPP^+ was a more potent neurotoxin for the cultured dopamine (DA) neurons as determined by decrease in [^{3}H]DA uptake by the cultures and decrease in endogenous levels of DA and homovanillic acid. Catecholamine histofluorescence demonstrated the loss of fluorescing fibers after exposure to MPP^+. The reduced analogue of MPTP, namely 1-methyl-4-phenylpiperidine, had no significant neurotoxic action on the dopaminergic neurons in culture.

1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) is a potent neurotoxin that destroys specifically the dopaminergic neurons of the substantia nigra (SN) zona compacta in humans [5] and in experimental animals [1, 6, 8]. Although rats are relatively insensitive to MPTP, we [12] have recently shown that organotypic cultures of embryonic rat SN neurons provide a useful model for studies of the toxic action of MPTP. Addition of micromolar concentrations of MPTP to the feeding medium causes selective destruction of the dopaminergic neurons in culture. Toxicity can be prevented by treating the cultures with the monoamine oxidase (MAO) inhibitors pargyline or deprenyl [3, 12]. Protection against MPTP neurotoxicity by MAO inhibitors was subsequently reported in monkeys [4, 9] and mice [7].

MPTP can be metabolized by brain mitochondrial MAO to the 1-methyl-4-phenylpyridinium ion (MPP^+) [2]. MPP^+ has been isolated from the brain of animals treated with MPTP [10, 11] and its accumulation in the brain was inhibited when the animals received a MAO inhibitor prior to the injection of MPTP. A dihydropyridine intermediate of MPTP metabolism, as well as MPP^+, and hydrogen peroxide plus associated oxy-radicals (all derived from MPTP and/or MAO activity) have been considered as potential candidates for the cellular toxin [2, 11, 12].

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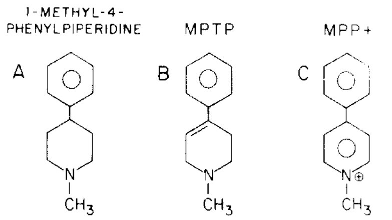

Fig. 1. Structure of 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP; B), 1-methyl-4-phenylpiperidine (reduced MPTP; A) and 1-methyl-4-phenylpyridinium ion (MPP^+, fully oxidized MPTP; C).

In the present communication, we report that MPP^+ (fully oxidized MPTP) is toxic to the dopaminergic neurons of rat midbrain in organotypic culture, while the reduced compound, 1-methyl-4-phenylpiperidine, has no neurotoxic action at similar concentrations (see Fig. 1A–C for chemical structures of the compounds). MPP^+ proved to be a more effective neurotoxin than MPTP in organotypic culture.

Organotypic cultures of ventral midbrain from rat embryos (16th gestational day) were set on collagen-coated coverslips in Maximow chambers. The feeding medium consisted of 25% rat serum (prepared in our laboratory), 10% chick embryo extract and 65% MEM with glutamine (GIBCO) supplemented with 600 mg glucose/100 ml. The cultures were left undisturbed for the first 8 days in vitro and, thereafter, were fed twice weekly. The drugs were added to the feeding medium. During the experimental period, cultures were fed with 75 μl of medium. The experiments were initiated on the 8th day in vitro. MPP^+-iodide and the piperidine analogue (HCl salt) were gifts from Dr. S.P. Markey (NIMH, Bethesda, MD) and MPTP-HCl was purchased from Research Biochemicals, Wayland, MA.

Living cultures were monitored with an inverted phase microscope for signs of generalized drug toxicity. The presence of dopaminergic axons in the cultures was documented by catecholamine (CA) histofluorescence with the glyoxylic acid technique [13]. In order to fully visualize the dopaminergic fibers, the cultures were incubated with 10 μM α-methylnorepinephrine for 30 min at 37°C before treatment with glyoxylic acid. A more quantitative estimate of the dopaminergic fibers and terminals present in the cultures was obtained by measuring the uptake of [^{3}H]dopamine ([^{3}H]DA; New England Nuclear; spec.act. 24.5–26.8 Ci/mmol) by individual cultures at 37°C for 10 min.

Measurements for homovanillic acid (HVA) and DA in protein-free extracts were carried out by high-performance liquid chromatography (Hewlett-Packard Model 1084B) with electrochemical detection (Bioanalytical Systems; set at +0.8 V). For DA assays, two cultures were combined and homogenized in 200 μl of 0.4 N perchloric acid containing α-methyldopamine (0.1 μg/ml) as internal standard; a μBondapak C_{18} column (Waters, 10-μm bead, 25 cm length) was used, and the solvent consisted of 50 mM acetate buffer, pH 5.1, containing 1 mM ethylenediaminetetraacetic acid (EDTA), 2.5 mM sodium octylsulfate and 20% (v/v) methanol. For HVA assays, 40-μl aliquots of feeding medium from each of two cultures were mixed with 320 μl 0.4 N perchloric acid containing 3-methoxy-4-hydroxy-phenylpropionic acid (final concn. 0.1 μg/ml; gift from Dr. Y.H. Lowe, Dept. Neurology, Mount Sinai School of Medicine) as internal standard; an Ultrasphere column (Altex, 5-μm bead, 15 cm length) was used, and the solvent consisted of 50 mM phosphate buffer, pH 3.1, containing 1 mM EDTA and 12% (v/v) acetonitrile. Samples were centrifuged in the cold for 2.5 min on a Fisher Micro-Centrifuge (Model 235B) prior to assay.

The morphological appearance of the fiber outgrowth around the explants was not affected by exposure to MPP^+ (10 μM for 7 days) as can be seen by comparing the phase contrast photomicrographs of a control and an MPP^+-treated culture (Fig. 2A, B). However, when the cultures were treated with glyoxylic acid for the histochemical visualization of CAs, there was a complete absence of fluorescing fibers in the axonal outgrowth of the MPP^+-treated cultures (Fig. 2C, D). Since incubation with α-methylnorepinephrine failed to induce fluorescence in the fibers of the MPP^+-treated cultures, the most likely interpretation is that the fibers had degenerated.

Destruction of the dopaminergic neurons after exposure to 10 μM MPP^+ for

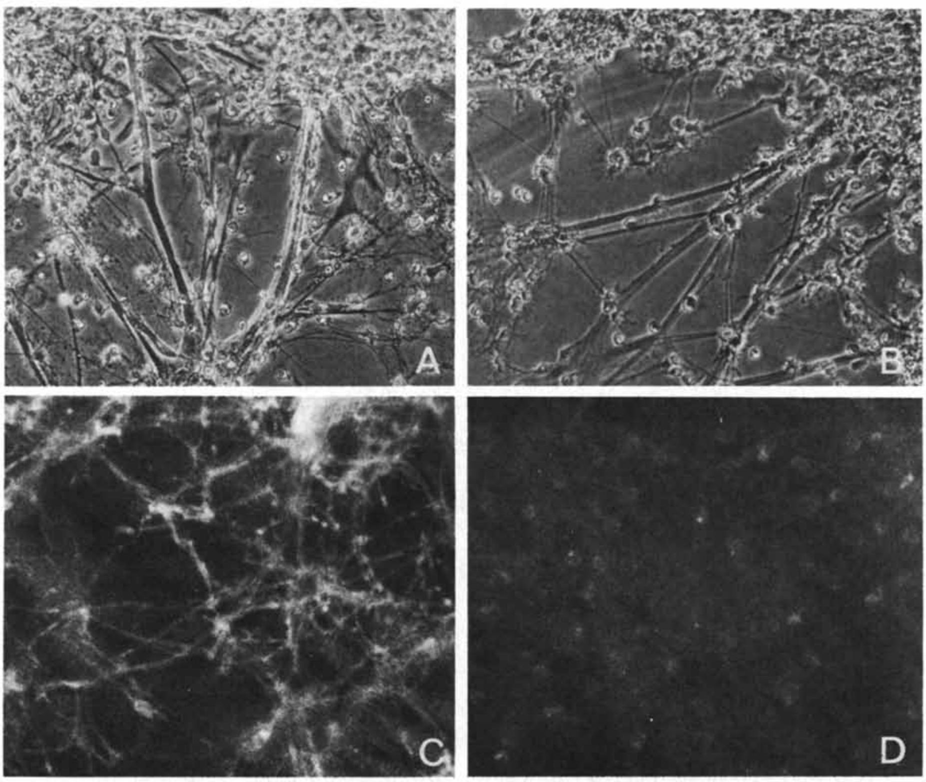

Fig. 2. A and B: phase-contrast photomicrographs of living cultures. Nerve fiber bundles are present in both the control culture (A) and the culture treated with 10 μM MPP^+ for 7 days (B). $\times 320$. C and D. Fluorescence photomicrographs of the same cultures after treatment for catecholamine (CA) histofluorescence. Note complete absence of fluorescing fibers from the MPP^+-treated culture (D). A prominent fluorescing plexus is present in the control culture (C). Both cultures were incubated with 10 μM α-methylnorepinephrine for 30 min before processing for CA histofluorescence. Magnification $\times 400$.

TABLE I

UPTAKE OF [^{3}H]DA BY CULTURES OF EMBRYONIC RAT VENTRAL MIDBRAIN ON THE 16th DAY IN VITRO

Exposure to the drugs (10 μM) began on the 8th day in vitro and lasted for 7 days. Drugs were withdrawn 24 h prior to [^{3}H]DA uptake. The values represent the pmoles [^{3}H]DA accumulated by individual cultures after exposure to 20 nM [^{3}H]DA for 10 min, followed by 10 min wash. *Differs from control, $P<0.05$; ** differs from control, $P<0.005$. Student's t-test.

Treatment	[^{3}H]DA uptake ($\pm$ S.E.M.)	% control	n
Control	0.74 ± 0.16	100	5
MPTP	0.25 ± 0.03*	34	4
MPP$^+$	0.03 ± 0.01**	4	5
1-Methyl-4-phenylpiperidine	0.58 ± 0.07	78	5

7 days was confirmed by measuring the decrement in high-affinity uptake of [^{3}H]DA by the cultures. The effect of MPP$^+$ was compared to the effect of the same concentration of MPTP or the piperidine analogue. The results are shown in Table I. The uptake of [^{3}H]DA at 24 h after withdrawal of drug was only 4% of the control value for MPP$^+$-treated cultures and 34% of control for MPTP-treated cultures. Treatment of the cultures with the piperidine analogue did not significantly affect the uptake of [^{3}H]DA. A recent study [10] has shown that 1-methyl-4-phenylpiperidine is not cytotoxic in monkeys.

At a concentration of 10 μM, MPP$^+$ causes significant damage to the dopaminergic neurons in the culture even within 24 h. After 24 h of exposure to MPP$^+$ and subsequent washing for another 24 h, the uptake of [^{3}H]DA was reduced to 46% of the control levels (pmol DA/culture: control 0.16 ± 0.02, n = 5, MPP$^+$ 0.07 ± 0.01, n = 5; MPP$^+$ differs from control $P<0.01$). Under the same conditions MPTP had no toxic effect on the DA neurons in the cultures (pmol DA/culture: control 0.29 ± 0.03, n = 4; MPTP 0.25 ± 0.04, n = 4, $P>0.1$).

In order to determine whether or not the effect of MPP$^+$ on the uptake of [^{3}H]DA could be reversed by prolonged washing, cultures were treated with 10 μM MPP$^+$ for 7 days, and then they were fed twice with normal feeding medium (without MPP$^+$) and finally analysed for [^{3}H]DA uptake at 5 days after withdrawal of MPP$^+$. The [^{3}H]DA uptake of the MPP$^+$-treated cultures was only 8% of the control value (n = 2 per group). This result indicates that the effect of MPP$^+$ is not reversed by prolonged washing of the cultures.

The toxicity of MPP$^+$ and MPTP increased with increasing concentration of the drugs (Table II). The sharp cut-off point for each drug gives an indication of a critical concentration of drug necessary to initiate cellular toxicity. MPP$^+$ was more potent than MPTP. At a concentration of 2.5 μM, MPP$^+$ destroyed 65% of the DA fibers, whereas the same concentration of MPTP was without any effect. The greater toxicity of MPP$^+$ (Table II) correlates with the greater damage observed for 10 μM MPP$^+$ compared to 10 μM MPTP after 24 h exposure (see above).

The levels of HVA accumulated in the feeding medium and the endogenous levels

TABLE II

EFFECT OF DIFFERENT CONCENTRATIONS OF MPP^+ AND MPTP ON THE UPTAKE OF $[^3H]$DA BY CULTURES OF EMBRYONIC RAT MIDBRAIN

Cultures were exposed to the drugs for 7 days and the uptake was measured 24 h after removal of the drugs. Uptake of control cultures for MPP^+ was 0.52 ± 0.03 and for MPTP 0.24 ± 0.03, pmol $[^3H]$DA/culture.*Differs from control, $P < 0.005$; Student's t-test.

Drug	Concentration (μM)	% control	n
MPP^+	0.2	100±15	6
	0.6	110±21	5
	2.5	35± 4*	6
	10.0	5± 0.4*	5
MPTP	2.5	100±14	6
	10.0	44± 9*	6
	25.0	42± 4*	7

of DA in the cultures were greatly reduced by exposure to MPP^+. Three days after withdrawal of the drug (10 μM MPP^+ for 7 days), the HVA levels in the feeding medium were reduced to 24% of control levels (control 37.1 ± 10.5 and MPP^+ 8.7 ± 0.7 ng HVA/75 μl of medium used to feed a single culture, $P < 0.05$), while the levels of endogenous DA in the cultures dropped from 7.1 ± 2.2 to 0.35 ± 0.1 ng/culture ($n = 7$, $P < 0.05$).

Our results indicate that MPP^+ is a potent neurotoxin for DA neurons in culture, whereas 1-methyl-4-phenylpiperidine is not. MPP^+ causes destruction of the DA fibers and prolonged depletion of DA and HVA levels. Its toxic effects are produced at shorter time intervals and lower concentrations than MPTP. These observations offer support for the idea [11] that MPP^+ could be responsible, in part, for the neurotoxicity of MPTP. However, our results do not exclude the possibility that other products of the metabolic activation of MPTP or other products of the enzymatic activity of MAO may be toxic to DA neurons.

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1 Burns, R.S., Chiueh, C.C., Markey, S.P., Ebert, M.H., Jacobowitz, D.M. and Kopin, I.J., A primate model of parkinsonism: selective destruction of dopaminergic neurons in the pars compacta of the substantia nigra by N-methyl-4-phenyl-1,2,3,6-tetrahydropyridine, Proc. Natl. Acad. Sci. USA, 80 (1983) 4546–4550.

2 Chiba, K., Trevor, A. and Castagnoli, N., Jr., Metabolism of the neurotoxic tertiary amine, MPTP, by brain monoamine oxidase, Biochem. Biophys. Res. Commun., 120 (1984) 574–578.

3 Cohen, G. and Mytilineou, C., Studies on the mechanism of action of 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP), Life Sci., 35 (1985) 237–242.

4 Cohen, G., Pasik, P., Cohen, B., Leist, A., Mytilineou, C. and Yahr, M.D., Pargyline and deprenyl prevent the neurotoxicity of 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) in monkeys, Europ. J. Pharmacol., 106 (1984) 209–210.

5 Davis, G.C., Williams, A.C., Markey, S.P., Ebert, M.H., Caine, E.D., Reichert, C.M. and Kopin, I.J., Chronic parkinsonism secondary to intravenous injection of meperidine analogues, Psychiat. Res., 1 (1979) 249–254.
6 Heikkila, R.E., Hess, A. and Duvoisin, R.C., Dopaminergic neurotoxicity of 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine in mice, Science, 224 (1984) 1451–1453.
7 Heikkila, R.E., Manzino, L., Cabbat, F.S. and Duvoisin, R.C., Protection against the dopaminergic neurotoxicity of 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine by monoamine oxidase inhibitors, Nature (Lond.), 311 (1984) 467–469.
8 Langston, W.J., Forno, L.S., Rebert, C.S. and Irwin, I., Selective nigral toxicity after systemic administration of 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine (MPTP) in the squirrel monkey, Brain Res., 292 (1984) 390–394.
9 Langston, J.W., Irwin, I., Langston, E.B. and Forno, L.S., Pargyline prevents MPTP-induced parkinsonism in primates, Science, 225 (1984) 1480–1482.
10 Langston, J.W., Irwin, I., Langston, E.B. and Forno, L.S., The importance of the '4–5' double bond for neurotoxicity in primates of the pyridine derivative MPTP, Neurosci. Lett., 50 (1984) 289–294.
11 Markey, S.P., Johannessen, J.N., Chiueh, C.C., Burns, R.S. and Herkenham, M.A., Intraneuronal generation of a pyridinium metabolite may cause drug-induced parkinsonism, Nature (Lond.), 311 (1984) 464–467.
12 Mytilineou, C. and Cohen, G.,1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine destroys dopamine neurons in explants of rat embryo mesencephalon, Science, 225 (1984) 529–531.
13 Mytilineou, C., Cohen, G., Dembiec-Cohen, D., Van Woert, M. and Hwang, E., Explant cultures of dog substantia nigra and striatum: a model for the study of nigro-striatal dopamine neurons, J. Neural Trans., Suppl. 19 (1983) 37–51.